

Surface Roughness Modelling and Field-Based Conductivity Extraction for High-Frequency Waveguide Structures Using HFSS

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Abstract—Accurate modelling of conductor surface roughness is essential in predicting ohmic loss for high-frequency waveguide structures such as W-band Folded Waveguide Traveling-Wave Tubes (FWTWTs). At millimetre-wave frequencies, skin depth becomes comparable to machined surface asperities, significantly increasing conductor loss. This work develops a validated HFSS methodology using Groiss and Huray roughness models and introduces a general field-based extraction technique for determining surface resistance (R_s) and effective conductivity (σ_{eff}). Custom HFSS Field Calculator expressions were formulated to compute surface current density and surface loss integrals. Results confirm that roughness minimally affects field distribution but significantly degrades surface impedance, validating the proposed extraction workflow for future high-power vacuum electronic device design.

Index Terms—Surface roughness, Groiss model, Huray model, HFSS, surface resistance, effective conductivity, folded waveguide TWT.

I. INTRODUCTION

High-frequency electromagnetic structures such as W-band Folded Waveguide Traveling-Wave Tubes (FWTWTs) require accurate estimation of conductor loss. As frequency increases, the skin depth of brass and copper reduces to sub-micron values, making surface roughness a dominant contributor to attenuation.

Classical techniques for measuring complex conductivity—cavity perturbation, reflection-based methods, or transmission-line attenuation—are not directly applicable to complex slow-wave structures. Hence, a simulation-driven approach using full-wave electromagnetic solvers becomes essential.

This work achieves the following:

- Develops an HFSS-based simulation workflow for modelling roughness in rectangular waveguides and folded waveguide TWT structures.
- Implements and compares Groiss and Huray roughness models.
- Creates HFSS Field Calculator expressions for surface current density and surface loss.
- Introduces a field-integral-based extraction method for surface resistance R_s and effective conductivity σ_{eff} .

- Validates the independence of modal fields from roughness.

II. HFSS MODELLING OF W-BAND FWTWT STRUCTURE

A folded-waveguide slow-wave section was constructed in HFSS using brass as the conductor. A Finite Conductivity boundary with parametric RMS roughness S_r was applied only to the internal metallic surfaces, reflecting real machined conditions.

A. Geometry and Boundary Conditions

The model includes input/output ports, periodic waveguide folds, and adaptive meshing suitable for narrow W-band features.

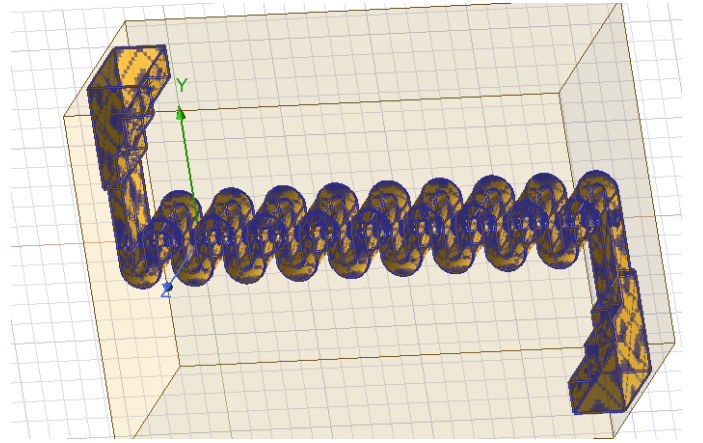


Fig. 1: HFSS model and boundary assignments.

B. S-Parameter Verification

Simulated S-parameters were compared to reference results from similar past designs to confirm correctness of the geometry and boundary setup.

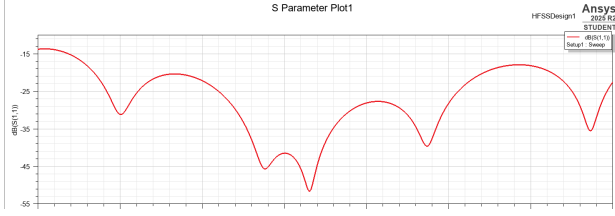


Fig. 2: Simulated S-parameters

III. SURFACE ROUGHNESS MODELLING

A. Skin depth for brass at 10.625 GHz

We compute the skin depth to justify the chosen roughness sweep range. Using

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}},$$

with $f = 10.625 \times 10^9$ Hz, $\mu_0 = 4\pi \times 10^{-7}$ H/m, and $\sigma = 1.5 \times 10^7$ S/m (brass), we obtain

$$\pi f \mu_0 \sigma \approx 6.295 \times 10^{11},$$

$$\delta = \frac{1}{\sqrt{6.295 \times 10^{11}}} \approx 1.26 \times 10^{-6} \text{ m} = 1.26 \text{ } \mu\text{m}.$$

This confirms that a roughness sweep up to $1 \text{ } \mu\text{m}$ and Huray nodule radii up to $0.8 \text{ } \mu\text{m}$ span the sub-skin, comparable-skin and above-skin regimes and are therefore physically relevant.

B. Groiss Model

The Groiss model modifies the smooth-surface resistance using:

$$\text{fact}_R = 1 + \exp \left[- \left(\frac{\delta}{2s} \right)^{1.6} \right],$$

where δ is the skin depth and s is RMS roughness height. The model is valid for moderate roughness levels (typically $\text{fact}_R \leq 2$).

C. Huray Model

The Huray “snowball” model uses nodule radius r and surface coverage density ρ . In our implementation we sweep:

$$r \in [0, 0.8] \text{ } \mu\text{m}, \quad \rho \in \{0.25, 0.50\},$$

to capture sparse and moderate coverage representative of machined surfaces.

$$R_s^{\text{Huray}} = R_s^{\text{smooth}} \left(1 + \frac{2\pi r^2 \rho}{\delta} \right).$$

Huray generally predicts stronger loss scaling and is more physically relevant for machined or etched surfaces.

IV. HFSS SIMULATION SETUP

Roughness parametric studies were performed on a simple rectangular waveguide test section (TE_{10}) to isolate roughness physics and allow dense parameter sweeps. The full FWTWT geometry was simulated separately for device-level S-parameter verification.

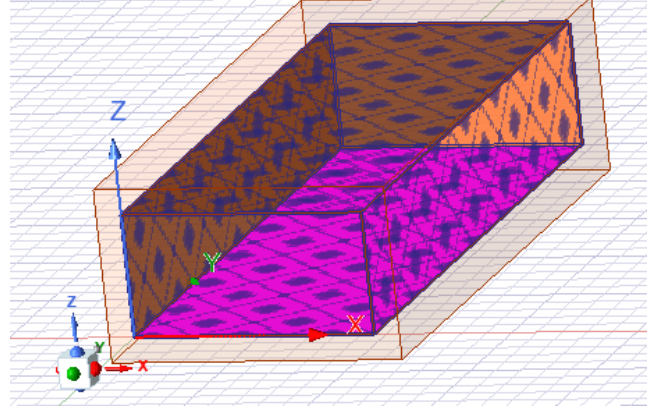


Fig. 3: Rectangular waveguide test-section used for roughness extraction. **Bottom_layer** (highlighted) is the surface used for surface loss and current integrals.

Finite-conductivity boundaries with Groiss or Huray parameters were assigned to metallic walls. A discrete-frequency solution point was used (fields exported at the chosen frequency). Surface integrals were evaluated on the Bottom_layer surface.

V. SURFACE CURRENT, SURFACE LOSS AND CONDUCTIVITY EXTRACTION

The phase-corrected surface current magnitude used in our analysis is defined as

$$J_s(\mathbf{r}) = \left| \text{AtPhase}(\langle J_x, J_y, J_z \rangle, \text{Var}(\text{Phase})) \right|,$$

and the reported scalar is the maximum value over the selected surface:

$$J_{s,\text{max}} = \max_{\mathbf{r} \in S} J_s(\mathbf{r}).$$

$J_{s,\text{max}}$ was evaluated on the Bottom_layer for each roughness case to validate whether roughness alters the modal currents. Results (Fig. 4) show negligible variation of $J_{s,\text{max}}$ with roughness, confirming that roughness primarily affects loss (surface impedance), not modal field distribution.

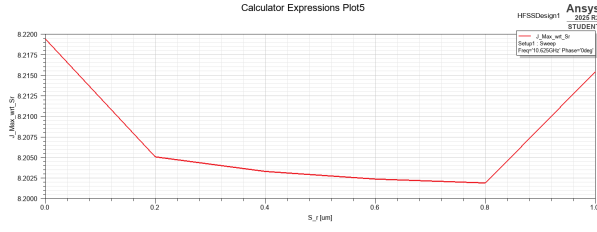
The integrated surface loss on Bottom_layer is computed as

$$I_{\text{SLD}} = \int_S \text{SurfaceLossDensity}(\mathbf{r}) dS.$$

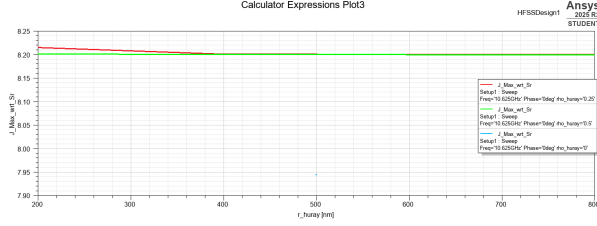
Since modal fields remain essentially unchanged, conductor loss increases monotonically with S_r or Huray parameters. From the simulated fields we compute an effective surface resistance

$$R_s = 2 \cdot \frac{\int_S \text{SLD} dS}{\int_S |J_s|^2 dS},$$

which is independent of port-power normalization.

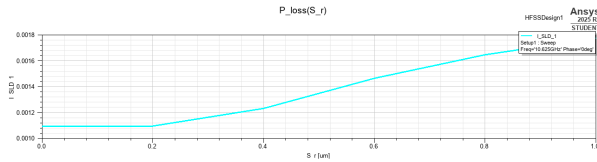


(a) Groiss: $J_{s,\max}$ vs S_r .

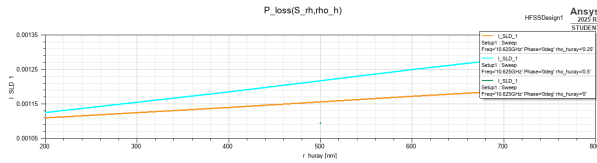


(b) Huray: $J_{s,\max}$ vs nodule radius (for $\rho = 0.25, 0.5$).

Fig. 4: Phase-corrected surface-current maxima on Bot-
tom_layer for Groiss and Huray models.



(a) Groiss: Integrated surface loss I_{SLD} vs S_r .



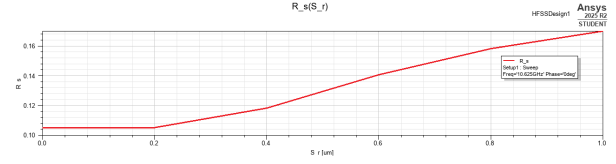
(b) Huray: Integrated surface loss I_{SLD} vs Huray parameters.

Fig. 5: Integrated surface loss on Bottom_layer for Groiss and Huray models.

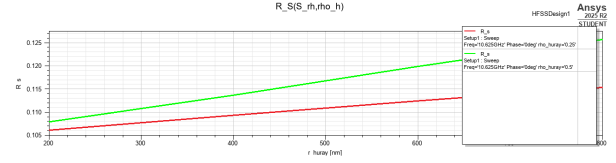
From the extracted R_s we compute the effective conductivity

$$\sigma_{\text{eff}} = \frac{\pi f \mu_0}{R_s^2},$$

the apparent conductivity seen by the guided mode (not the DC bulk). Both R_s and σ_{eff} trends are plotted in Figs. 6 and 7.

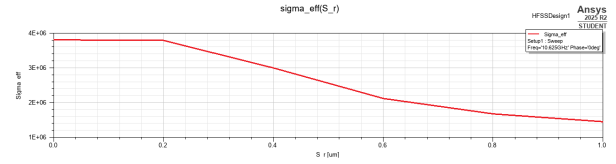


(a) Groiss: R_s vs S_r .

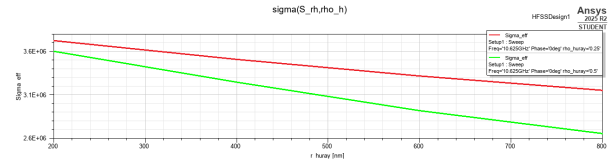


(b) Huray: R_s vs Huray parameters.

Fig. 6: Extracted surface resistance R_s from field integrals.



(a) Groiss: σ_{eff} vs S_r .



(b) Huray: σ_{eff} vs Huray parameters.

Fig. 7: Effective conductivity σ_{eff} trends for Groiss and Huray.

VI. RESULT SUMMARY

The roughness-parametric HFSS study demonstrates that surface roughness affects conductor losses and effective surface impedance, while leaving the modal field distribution essentially unchanged.

A. Surface Current Behaviour

Across all Groiss and Huray roughness sweeps, the maximum phase-corrected surface current $J_{s,\max}$ on the analyzed waveguide wall remains nearly constant. This confirms that roughness modifies only the loss boundary condition and does not alter the electromagnetic mode structure of the TE₁₀ guide used for extraction.

B. Surface Loss Trends

The integrated surface-loss density I_{SLD} increases monotonically with roughness for both models. Groiss predicts a smooth, gradual rise with S_r , while Huray exhibits a stronger and parameter-dependent increase, reflecting its more detailed micro-geometry representation.

C. Extracted Surface Resistance

The extracted surface resistance,

$$R_s = 2 \frac{\int_S \text{SLD } dS}{\int_S |J_s|^2 dS},$$

increases systematically with roughness, consistent with classical roughness-induced impedance enhancement. Huray produces a higher sensitivity to particle radius and surface coverage than Groiss, indicating a more realistic treatment for moderately to highly rough surfaces.

D. Effective Conductivity

Using the extracted R_s , the effective high-frequency conductivity is obtained as

$$\sigma_{\text{eff}} = \frac{\pi f \mu_0}{R_s^2}.$$

The resulting σ_{eff} decreases with increasing roughness. Both models predict degradation relative to bulk conductivity, with Huray showing a larger reduction due to its physical basis in surface asperity geometry.

E. Overall Observations

- Roughness does not significantly modify the modal current distribution but directly impacts conductor loss.
- I_{SLD} , R_s , and σ_{eff} exhibit monotonic trends reflecting increased dissipation with roughness.
- Huray provides a more sensitive and physically grounded representation of rough surfaces, whereas Groiss offers a simpler correction suitable for smooth-to-moderate roughness.

These results establish a complete HFSS-based workflow for extracting roughness-dependent loss parameters, forming a basis for future comparison with metrology-derived roughness measurements and experimental attenuation data.

VII. ASSUMPTIONS AND LIMITATIONS

The following assumptions/limitations apply:

- Roughness is modelled via Groiss/Huray boundary models (statistical/microgeometry approximations); microtopography is not explicitly meshed.
- Rectangular test-section assumes isotropic roughness on Bottom_layer; side-wall differences are not included in these runs.
- Linear small-signal EM; no thermal or nonlinear high-power effects considered.
- Mesh convergence should be checked for publication-quality numerical values.

VIII. FUTURE WORK

Planned next steps:

- 1) **Metrology-driven modelling:** Acquire profilometer/interferometer scans of machined waveguides to extract actual roughness statistics (RMS, autocorrelation length, nodule radii) and use them to drive Huray parameters.

- 2) **Geometry-aware roughness:** Update the FWTWT model to apply distinct roughness parameters on bottom and side walls based on machining tool-path analysis.
- 3) **Fabrication and experimental validation:** Fabricate test sections, measure insertion loss, invert to experimental R_s , and fit Huray parameters to measured data.
- 4) **High-power/thermal effects:** Extend the model to include temperature-dependent conductivity and thermal hotspots for high-power operation.

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